

Hon Hao Yuan (Brandon)

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EDUCATION

National University of Singapore (NUS)

Aug 2025 – May 2026

Non-Graduating Non-Exchange Programme (NGNE), School of Computing

- Admitted into a highly selective immersion programme for international undergraduates.
- Executed a faculty-supervised Final Year Project (FYP) focused on engineering a real-time observability architecture for Software-Defined Networks (SDNs).

Beijing Institute of Technology (BIT)

Aug 2022 – June 2026

Computer Science and Technology, School of Computing

- **GPA:** 3.67 / 4.0 | **Weighted Average Mark:** 89.05 / 100.00

EXPERIENCE

Part-Time Software Engineering Intern | NUS-NCS Joint Lab for Cyber Security

Sep 2025 – Nov 2025

- Engineered a **secure** and **reliable** automated **Python**-based data ingestion pipeline, streamlining the extraction and mapping of data into a **Neo4j** graph database.
- Analyzed system architecture to evaluate and resolve **performance bottlenecks**, improving overall data processing throughput and **infrastructure scalability**.
- Facilitated daily communication and cross-team coordination with researchers to validate data models and **triage** potential integration issues.

Research Assistant | Beijing Institute of Technology

Nov 2023 – May 2024

- Analyzed academic publication data using multithreaded **Python** scripts and managed **Elasticsearch** indices for data exploration.

PROJECTS

Observable Global Control Plane for SDNs (FYP) | GitHub

Dec 2025 – May 2026

Advised by Assoc. Prof. Ma Tianbai (Richard)

Go, Kubernetes, Docker, Kafka, OpenTelemetry, Grafana

- Co-architected a distributed Software-Defined Network (SDN) control plane, building the microservices using **Golang**, packaging into **Docker** containers under **Ubuntu Linux**, and managing **deployment** onto a local **Kubernetes (k8s)** cluster.
- Spearheaded system **observability** by deeply integrating **OpenTelemetry**, ensuring the network architecture remains **reliable, scalable, and secure**.
- Managed **infra** and provisioned scalable telemetry pipelines utilizing **Kafka**, creating comprehensive **monitoring and dashboarding** solutions via **Grafana** to visualize real-time network states.
- Engineered automated **issue triage** mechanisms using **Alertmanager** and custom metrics to trigger real-time Telegram alerts, proactively analyzing and responding to potential risks like soft failures.

AI-Driven Solana Market Scanner (Web3 Hackathon) | GitHub

June 2026 – June 2026

AI Agentic Workflows, Model Context Protocol (MCP), LLM Reasoning Pipelines

- Architected an automated scanning tool via a no-code **AI agentic workflow**, utilizing AI as an engineering copilot to translate written plans into actionable implementation checkpoints.

- Integrated **Bitget MCP tools** to provision a real-time data ingestion pipeline, actively **monitoring** live market telemetry, news feeds, and backtesting data for automated system analysis.
- Designed an evidence-first “bull, bear, and judge” automated evaluation flow, structuring **reliable** LLM reasoning pipelines for objective data analysis and automated risk evaluation.

ProgressArc | *GitHub*

Dec 2024 – Feb 2025

Vue.js, Python, Flask, MongoDB, REST APIs

- Engineered a scalable **Python Flask** backend with robust **REST APIs** for secure **login services** and registration.
- Designed a responsive **Vue.js** frontend and utilized **MongoDB** for data scalability, simplifying **infra management** and operations (**O&M**).

FoodBridge | *GitHub*

Feb 2024 – March 2024

React Native, Convex

- Developed a cross-platform **React Native** mobile app and integrated **Convex** as a plugged-in backend to manage real-time data synchronization, ensuring a **reliable** and **scalable** cloud infrastructure.

SKILLS

- **Programming & Scripting:** Python, Go, Bash, SQL, C++, C
- **Cloud, DevOps & O&M:** AWS, Kubernetes, Docker, Helm, Kafka, Prometheus, Grafana, OpenTelemetry, Linux
- **Backend, AI & Architecture:** Microservices, Distributed Systems, REST APIs, Flask, AI Agents, MCP, Convex
- **Databases:** Elasticsearch, MongoDB, MySQL, Neo4j
- **Language Skills:** Bilingual proficiency in Chinese and English